8

Inferences About Proportions

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Several inferential tests are covered in this chapter, depending on (a) whether there are one or two samples, (b) whether the two samples are selected in an independent or dependent manner, and (c) whether there are one or more categorical variables. More specifically, the topics described include the following inferential tests: testing whether a single proportion is different from a hypothesized value; testing whether two independent proportions are different; testing whether two dependent proportions are different; and the chi-square goodness-of-fit test and chi-square test of association. We use many of the foundational concepts previously covered in Chapters 6 and 7. New concepts to be discussed include the following: proportion; sampling distribution and standard error of a proportion; contingency table; chi-square distribution; and observed versus expected frequencies. Our objectives are that by the end of this chapter, you will be able to (a) understand the basic concepts underlying tests of proportions, (b) select the appropriate test, and (c) determine and interpret the results from the appropriate test.

8.1 What Inferences About Proportions Involving the Normal Distribution Are and How They Work

A superbly talented set of four graduate students have been expertly completing research projects through their work in the statistics lab. We find the group, once again, ready for a challenge!

The statistics lab has been contracted to work with Dr. Senata, the Director of the Undergraduate Services Office at Ivy Covered University, and Dr. Walnut, a lobbyist from a state that is considering legalizing gambling. Challie Lenge will be advising Dr. Senata, and Addie Venture will be working with Dr. Walnut.

In conversation with Challie, Dr. Senata shares that she recently read a report that provided national statistics on the proportion of students that major in various disciplines. Dr. Senata wants to know if there are similar proportions at their institution. Dr. Senata suggests the following research question: *Are the sample proportions of undergraduate student college majors at Ivy Covered University in the same proportions of those nationally?* Challie suggests a chi-square goodness of fit test as the test of inference. Her task is then to assist Dr. Senata in generating the test of inference to answer her research question.

Addie is consulting with Dr. Walnut, a lobbyist who is lobbying against legalizing gambling in his state. Dr. Walnut wants to determine if there is a relationship between level of education and stance on a proposed gambling amendment. Addie suspects that the proportions supporting gambling vary as a function of their education level. The following research question is suggested by Addie: *Is there an association between level of education and stance on gambling?* Addie suggests a chi-square test of association as the test of inference. Her task is then to assist Dr. Walnut in generating the test of inference to answer the research question.

This section deals with concepts and procedures for testing inferences about proportions that involve the normal distribution. Following a discussion of the concepts related to tests

Working in **R**, we can compute the effect size *h* using the *pwr* package.

```
install.packages("pwr")
library(pwr)
```

With the *install.packages* function, we install the *pwr* package. Using the *library* function, we then load *pwr* into our library.

```
h<-ES.h(0.40,.65)
h
```

We use the *ES.h* function and include our proportions in parentheses, where the first value represents p_1 and the second value represents p_2 Using the data from the headache example, our proportions are .40 and .65. From the results, we create an object named "h." The second line of script is simply telling **R** to output the results, which we see here. Thus, $h = -.51$. Using Cohen's conventions, this represents a moderate effect.

```
[1] -0.5060506
```

```
pwr.p.test(h=h,n=100,sig.level=0.05,
alternative="two.sided")
```

The *pwr.p.test* function can be used to compute observed power given the observed *h*, sample size (i.e., "n=100"), alpha level ("sig.level=0.05"), and two-sided test (alternative = "two.sided"). In this example, we find power to be .999, which indicates very high power.

```
proportion power calculation for binomial distribution (arcsine transformation)

           h = 0.5060506
           n = 100
   sig.level = 0.05
       power = 0.9990342
 alternative = two.sided
```

FIGURE 8.1
Computing effect size *h* in **R**.

8.1.4 Assumptions

For inferences about proportions assuming the normal distribution, it is assumed that the sample (in the case of a single proportion) or samples (in the case of independent and dependent proportions) have been randomly selected from the population (i.e., the assumption of independence) and that the normal distribution is the appropriate sampling distribution.

8.2 What Inferences About Proportions Involving the Chi-Square Distribution Are and How They Work

This section deals with concepts and procedures for testing inferences about proportions that involve the chi-square distribution. Following a discussion of the chi-square distribution relevant to tests of proportions, inferential tests are presented for the chi-square goodness-of-fit test and the chi-square test of association.

8.2.3.2 Chi-Square Test of Association Effect Size

Several measures of effect size, such as correlation coefficients and measures of association, can be requested in SPSS or computed in **R**, and are commonly reported effect size indices for results from chi-square tests of association. Which effect size value is selected depends in part on the measurement scale of the variable. For example, researchers working with nominal data can select a contingency coefficient: phi (for 2 × 2 tables), Cramer's V (for tables larger than 2 × 2), lambda, or an uncertainty coefficient. Correlation options available for ordinal data include gamma, Somer's d, Kendall's tau-b, and Kendall's tau-c. From the contingency coefficient, C, we can compute Cohen's w as follows:

$$w = \sqrt{\frac{C^2}{1-C^2}}$$

Cohen's recommended subjective standard for interpreting w (as well as the other correlation coefficients presented) is as follows: small effect size, w = .10, medium effect size, w = .30, large effect size, w = .50. See Cohen (1988) for further details. We will later review how to compute confidence intervals for w.

8.2.4 Assumptions

8.2.4.1 Chi-Square Goodness-of-Fit Assumptions

Two assumptions are made for the chi square goodness-of-fit test: (a) observations are *independent* (which is met when a random sample of the population is selected) and (b) an *expected* frequency of at least five per cell (and in the case of the chi-square goodness-of-fit test, this translates to an expected frequency of at least five per category, as there is only one variable included in the analysis). When the expected frequency is less than five, that particular cell (i.e., category) has undue influence on the chi-square statistic. In other words, the chi-square goodness-of-fit test becomes too sensitive when the expected values are less than five.

8.2.4.1 Chi-Square Test of Association Assumptions

The same two assumptions that apply to the chi-square goodness-of-fit test also apply to the chi-square test of association: (a) observations are independent (which is met when a random sample of the population is selected) and (b) an *expected* frequency of at least five per cell. When the expected frequency is less than five, that particular cell has undue influence on the chi-square statistic. In other words, the chi-square test of association becomes too sensitive when the expected values are less than five.

8.3 Computing Inferences About Proportions Involving the Chi-Square Distribution Using SPSS

Once again we consider the use of SPSS for the example datasets. Although SPSS does not have any of the z procedures described in the first part of this chapter, it is capable of conducting both of the chi-square procedures described.

8.4 Computing Inferences About Proportions Involving the Chi-Square Distribution Using R

We will illustrate computing both the chi-square goodness-of-fit and chi-square test of association using **R**.

8.4.1 The Chi-Square Goodness-of-Fit Test

Next we consider **R** for the chi-square goodness-of-fit test. The scripts are provided within the blocks with additional annotation to assist in understanding how the commands work. Should you want to write reminder notes and annotation to yourself as you write the commands in **R** (and we highly encourage doing so), remember that any text that follows a hashtag (i.e., #) is annotation only and not part of the **R** script. Thus, you can write annotations directly into **R** with hashtags. We encourage this practice so that when you call up the commands in the future, you'll understand what the various lines of code are doing. You may think you'll remember what you did. However, trust us. There is a good chance that you won't. Thus, consider it best practice when using **R** to annotate heavily!

8.4.1.1 Reading Data Into R

```
getwd()
```

R is always pointed to a directory on your computer. To find out which directory it is pointed to, run this "get working directory" function. We will assume that we need to change the working directory, and will use the next line of code to set the working directory to the desired path.

```
setwd("E:/Folder")
```

To set the working directory, use the *setwd* function and change what is in quotation marks here to your file location. Also, if you are copying the directory name, it will copy in slashes. You will need to change the slash (i.e., \) to a backslash (i.e., /). Note that you need your destination name within quotation marks in the parentheses.

```
Ch8_CollegeMajor <- read.csv("Ch8_CollegeMajor.csv")
```

The *read.csv* function reads our data into **R**. What's to the left of the <- will be what the data will be called in **R**. In this example, we're calling the R dataframe "Ch8_CollegeMajor." What's to the right of the <- tells **R** to find this particular .csv file. In this example, our file is called "Ch8_CollegeMajor.csv." Make sure the extension (i.e., .csv) is included in your script. Also note that the name of your file should be in quotation marks within the parentheses.

```
names(Ch8_CollegeMajor)
```

The *names* function will produce a list of variable names for each dataframe, as follows. This is a good check to make sure your data have been read in correctly.

FIGURE 8.11
Reading data into **R**.

8.5 Data Screening

Because it is a nonparametric procedure, fewer assumptions are associated with chi-square tests, and only one is actually tested when the procedure is generated (that being the assumption of expected frequencies). Examination of the assumptions for the examples have been provided in previous sections and presented with the statistical software computer output.

8.6 Power Using G*Power

A priori power can be determined using specialized software (e.g., Power and Precision, Ex-Sample, G*Power) or power tables (e.g., Cohen, 1988), as previously described. However, because standard statistical software does not provide power information for the results of the chi-square test of association just conducted, let us use G*Power to compute the post hoc power of our test.

8.6.1 Post Hoc Power for the Chi-Square Test of Association Using G*Power

The first thing that must be done when using G*Power for computing post hoc power is to select the correct test family. In our case, we conducted a chi-square test of association; therefore, the toggle button must be used to change the test family to χ^2 (see the screenshot in Figure 8.15). Next, we need to select the appropriate statistical test. We toggle to "Goodness-of-fit tests: Contingency tables." The "Type of power analysis" then needs to be selected. To compute post hoc power, we select "Post hoc: Compute achieved power—given α, sample size, and effect size."

The "Input Parameters" must then be specified. The first parameter is specification of the effect size w (this was computed by hand from the contingency coefficient and w = .408). The alpha level we tested at was .05, the sample size was 80, and the degrees of freedom were 3. Once the parameters are specified, simply click "Calculate" to generate the achieved power statistics.

The "Output Parameters" provide the relevant statistics given the input just specified. In this example, we were interested in determining post hoc power given a two-tailed test, with an observed effect size of .408, an alpha level of .05, and sample size of 80. Based on those criteria, the post hoc power was approximately .88. In other words, with a sample size of 80, testing at an alpha level of .05, and observing a moderate to large effect of .408, the power of our test was .88; thus, the probability of rejecting the null hypothesis when it is really false is 88%, which is very high power. Keep in mind that conducting power analysis *a priori* is recommended so that you avoid a situation where, post hoc, you find that the sample size was not sufficient to reach the desired level of power (given the observed effect size and alpha level).

8.7 Recommendations

Box 8.1 summarizes the tests reviewed in this chapter and the key points related to each (including the distribution involved and recommendations for when to use the test).

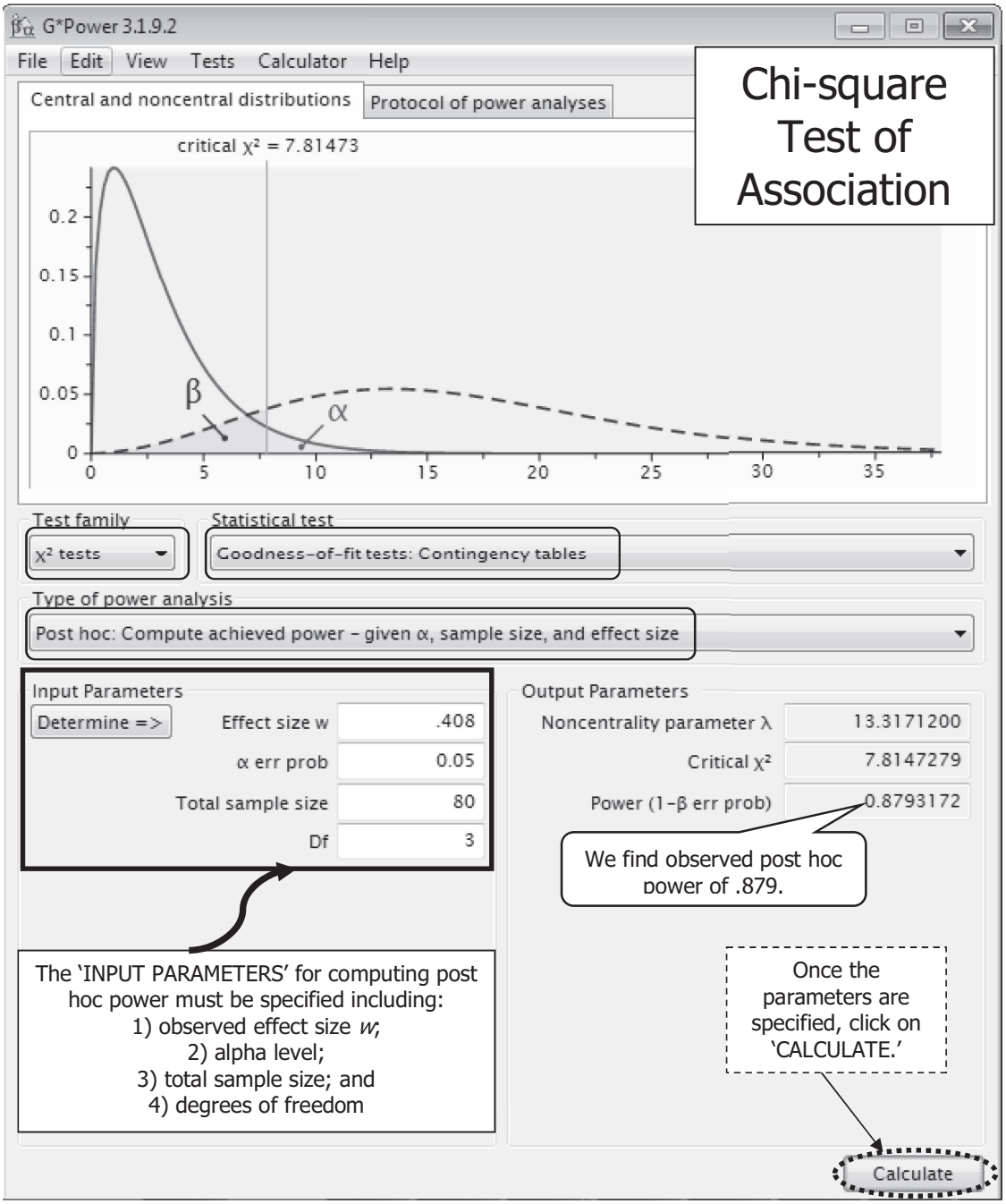


FIGURE 8.15
Chi-square test of association post hoc power.

BOX 8.1 Characteristics and Recommendations for Inferences About Proportions

Test	Distribution	When to Use
Inferences about a single proportion (akin to a one-sample mean test)	Unit normal, z	To determine if the sample proportion differs from a hypothesized proportion One variable, nominal or ordinal in scale
Inferences about two independent proportions (akin to the independent t test)	Unit normal, z	To determine if the population proportion for one group differs from the population proportion for a second independent group Two variables, both nominal or ordinal in scale
Inferences about two dependent proportions (akin to the dependent t test)	Unit normal, z	To determine if the population proportion for one group is different than the population proportion for a second dependent group Two variables of the same measure, both nominal or ordinal in scale
Chi-square goodness-of-fit test	Chi-square	To determine if observed proportions differ from what would be expected *a priori* One variable, nominal or ordinal in scale
Chi-square test of association	Chi-square	To determine association/relationship between two variables based on observed proportions Two variables, both nominal or ordinal in scale

8.8 Research Question Template and Example Write-Up

We finish the chapter by presenting templates and example write-ups for our examples. First we present an example paragraph detailing the results of the chi-square goodness-of-fit test and then follow this by the chi-square test of association.

8.8.1 Chi-Square Goodness-of-Fit Test

Recall that our graduate research assistant, Challie Lenge, was working with Dr. Senata, the Director of the Undergraduate Services Office at Ivy Covered University (ICU), to assist in analyzing the proportions of students enrolled in undergraduate majors. Challie's task was to assist Dr. Senata with writing her research question (*Are the sample proportions of undergraduate student college majors at ICU in the same proportions as those nationally?*) and generating the statistical test of inference to answer her question. Challie suggested a chi-square goodness-of-fit test as the test of inference. A template for writing a research question for a chi-square goodness-of-fit test follows:

> Are the sample proportions of [units in categories] in the same proportions of those [identify the source to which the comparison is being made]?

It may be helpful to include in the results of the chi-square goodness-of-fit test information on an examination of the extent to which the assumptions were met (recall there

A chi-square test of association was conducted to determine if there was a relationship between level of education and stance on gambling. The test was conducted using an alpha of .05. It was hypothesized that there was an association between the two variables. The assumption of an expected frequency of at least five per cell was met. The assumption of independence was not met because the respondents were not randomly selected; thus, there is an increased probability of a Type I error.

From Table 8.6 we can see from the row marginals that 55% of the individuals overall support gambling. However, lower levels of education have a much higher percentage of support, while the highest level of education has a much lower percentage of support. Thus, there appears to be an association or relationship between gambling stance and level of education. This is subsequently supported statistically from the chi-square test ($\chi^2 = 13.333$, $df = 3$, $p = .004$). Thus, the null hypothesis that there is no association between stance on gambling and level of education was rejected at the .05 level of significance. Examination of the standardized residuals suggests that respondents who hold a graduate degree are significantly more likely not to support gambling (standardized residual = 2.0) as compared to all other respondents. Further examination of column proportions using a Bonferroni-corrected z test suggests that individuals with less than a high school degree are statistically significantly different in proportions relative to all other education levels, proportions of those with a high school degree are also statistically different than undergraduate and graduate, and proportions of undergraduate and graduate also differ. The effect size, Cohen's w, was computed to be .408, which is interpreted to be a moderate to large effect (Cohen, 1988).

8.9 Additional Resources

In this chapter we described a third inferential testing situation, testing hypotheses about proportions. A number of resources have been provided throughout. For additional coverage of tests of proportion and analyzing categorical data, you may wish to consider Agresti (2013), among others. If you are interested in deeper coverage of chi-square, in particular, you may wish to consider Voinov, Balakrishnan, and Nikulin (2013) (which also includes an historical account and many other chi-square tests that are beyond the scope of this text) or Greenwood and Nikulin (1996).

Problems

Conceptual Problems

1. How many degrees of freedom are there in a 5 × 7 contingency table when the chi-square test of association is used?

a. 12

b. 24

c. 30

d. 35